Randev Ranjit

Digital Design & Embedded Systems | Incoming MSc CESE, TU Delft

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SUMMARY

Digital design engineer with synthesised + implemented FPGA designs on **Spartan-6** (Stump 16-bit RISC CPU, 98% slice occupancy, timing met) and **Artix-7** (RISC-V SoC + self-designed programmable cellular-automaton accelerator, live VGA demo). Production-scale Rust and C++ systems work; measured-parallelism on a 72-core NUMA system (30.7×). Incoming MSc CESE, TU Delft, Sept 2026.

EDUCATION

TU Delft MSc Computer & Embedded Systems Engineering (CESE) — Sept 2026 – 2028

University of Manchester BSc (Hons) Computer Science — 2023 – 2027 (expected)

Hardware modules: SoC Designs, Processor Microarchitecture, Multicore Computing, Computer Architecture, Microcontrollers, Digital Logic + Verilog.

HARDWARE & DIGITAL DESIGN

Programmable Cellular-Automaton SoC Accelerator *SystemVerilog, Vivado, Artix-7 XC7A50T* — 2025–26

- Designed and integrated a 1-D Wolfram-rule accelerator into a RISC-V SoC drawing engine; all 256 elementary rules runtime-selectable (8-bit register indexed by the 3-bit neighbourhood); 32-bit LFSR seed; `C_IDLE`→`C_RUN`→`C_DONE` FSM; byte-enabled framestore writes over a req/ack bus.
- Synthesised + implemented: whole-SoC 14,282 LUTs (43.8%), 7,226 FFs (11.1%), **72/75 BRAM (96%, framebuffer-bound)**; accelerator 4,433 LUTs / 1,467 FFs. Three-phase verification (unit, directed CA Rules 30/90/255, memory back-pressure with configurable `de_ack` stalls, full-system); demonstrated live on a monitor.

Stump 16-bit Multi-Cycle RISC Processor *Verilog, Xilinx ISE, Spartan-6 xc6slx4* — 2024

- Built a complete CPU from RTL primitives: datapath, 3-state FSM (FETCH/EXECUTE/MEMORY), 8-function ALU, shifter, sign-extender, 8×16-bit register file, instruction decode (ADD/ADC/SUB/SBC/AND/OR/LDST/BCC). Synthesised, mapped, placed and routed: **2,068/2,400 LUTs (86%), 591/600 occupied slices (98%), all signals routed, timing score 0.** Wrote Battleship in Stump assembly driving an MMIO 8×8 LED matrix.

LOW-LATENCY SYSTEMS SOFTWARE

ONI – Local LLM Coding Agent *Rust, 78 k LOC, single crate* — 2025–26

- Async Rust on `tokio`; ∼30 tools dispatched via fenced-block parser; family-specific parsers (Gemma/Qwen/MiniMax) with XML→canonical-fence normalisation. Process-group sandboxing via `process_group(0)` + `kill(-pid, SIGKILL)`. Criterion: parser < 100 μs/corpus, dispatch < 5 ms, loop-detector < 50 ns/KB. Cross-compiles pure-Rust to `x86_64-unknown-linux-musl`. Apple Silicon inference sweep: 44.4→19.3 tok/s vs. context depth, Metal memory arithmetic documented.

DAWgit – DAW Project Version Control *Rust daemon, 5 k LOC* — 2026

- macOS daemon: FSEvents → gzip-XML zero-copy parse (`roxmltree`) → semantic differ → `libgit2` per-file commit *without touching HEAD*; Unix-socket IPC with newline-delimited JSON, `tokio::select!` multiplexing; SHA-256 content-addressed audio LFS.

Driverless Racing Simulator (FSAI) *C++17, CMake, 34 k LOC* — 2024–25

- RK4 with adaptive sub-stepping; three YAML-switchable vehicle models (kinematic / single-track + Fiala / multi-body). Closed perception loop: OpenGL FBO stereo + async PBO readback → SIFT disparity → ONNX cone detector → Kalman map fusion. RAII nanosecond budget timers across sim/vision/control/CAN.

CONTROL, OPTIMISATION & PARALLELISM

Interpretable MPCC Racing Controller (Final-Year Dissertation) *Python, CasADi/IPOPT* — 2025–26

- N=20 MPCC OCP with b-spline track interpolants, GG friction-ellipse constraint, CasADi JIT-compiled NLP, dual warm-starting (primal + dual recycled), Radau IIA implicit plant integration. Benchmarked against RRHC + DDP baselines.

Multicore Computing – Measured Speedups *C/C++, pthreads, std::thread, OpenMP* — 2025

- pthreads vecadd, NUMA first-touch + affinity: **30.7× on 72 cores** (3% serial fraction). Jacobi stencil with `std::barrier` + temporal blocking (T_BLOCK=128) eliminates ∼90% of DRAM reads. Fine-spinlock dining philosophers: **1.04 M meals/s at 64 threads**, per-fork `pthread_spinlock_t` + lock-ordering.

Quant Trading System (side project) *Python, 40.9 k LOC, 81.9% coverage* — 2026

- Tick-level queue-position-aware HFT backtester (50 μs simulated order latency) + bar-level Nautilus; regime-gated multi-signal fusion (FinBERT/VADER/GDELT) under a 2-state Gaussian HMM Viterbi decoder; async SQLAlchemy on TimescaleDB; circuit-breaker risk controls.

INDUSTRY EXPERIENCE

AI / Software Engineering Intern *CarrLane Manufacturing, Chennai* — Jun – Aug 2025

Built an LLM + vector-search retrieval pipeline over a 10,000+ part engineering catalog; improved indexing and data quality for natural-language part selection.

TECHNICAL SKILLS

HDL & EDA: SystemVerilog, Verilog; Xilinx Vivado (synth + impl, Artix-7 & Spartan-7), Xilinx ISE (Spartan-6 PnR), ModelSim/Questa, SVA assertions, coverage-driven verification, register-mapped peripherals with interrupts.

Languages: Rust (production, 80 k+ LOC), C++17/20 (production, 45 k+ LOC), C, Python, Java, RISC-V & ARM assembly. **Numerics:** RK4 + adaptive sub-stepping, CasADi/IPOPT NLP, Kalman filtering, differential flatness.

Concurrency & Systems: pthreads, `std::thread`, OpenMP, atomics, barriers, spinlocks, NUMA affinity & first-touch, `tokio` async.